LAB REPORT 02

District Explorer: Building a Web Map from a Shapefile

Published URL: https://arshiabatool.github.io/webgis-538063-Arshia/Lab_02/

Repository Link: https://github.com/arshiabatool/webgis-538063-Arshia

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Q/A

1. Does the percentage difference grow or shrink as you move north?

The percentage difference generally grows as we move north. In our results, Thatta in the south has a difference of 20.24%, while Shangla farther north has a difference of 48.79%. Web Mercator increasingly exaggerates area at higher latitudes.

2. Compare the predicted inflation factor with the measured difference

The theoretical inflation factor is:

$$ F=\frac{1}{\cos^2(\phi)} $$

For Shangla, using an approximate latitude of 34.9°:

$$ F=\frac{1}{\cos^2(34.9^\circ)} \approx1.49 $$

The predicted inflation is therefore approximately:

$$ (1.49-1)\times100\approx49.1\% $$

Our measured difference is 48.79%, which is very close to the predicted value.

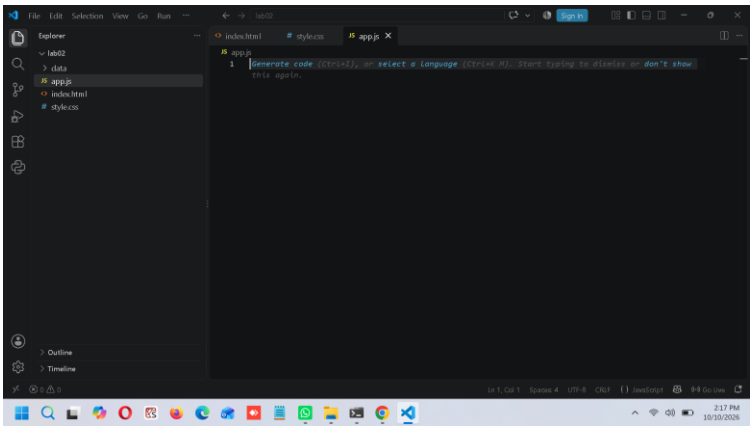
3. Which figure would you put in a report, and why?

I would use the UTM area because it is more suitable for local area measurements than Web Mercator. Web Mercator is designed mainly for displaying maps on the web and exaggerates area, especially farther from the Equator.

For rigorous reporting, the appropriate UTM zone or an equal-area coordinate system should be selected for the study area.

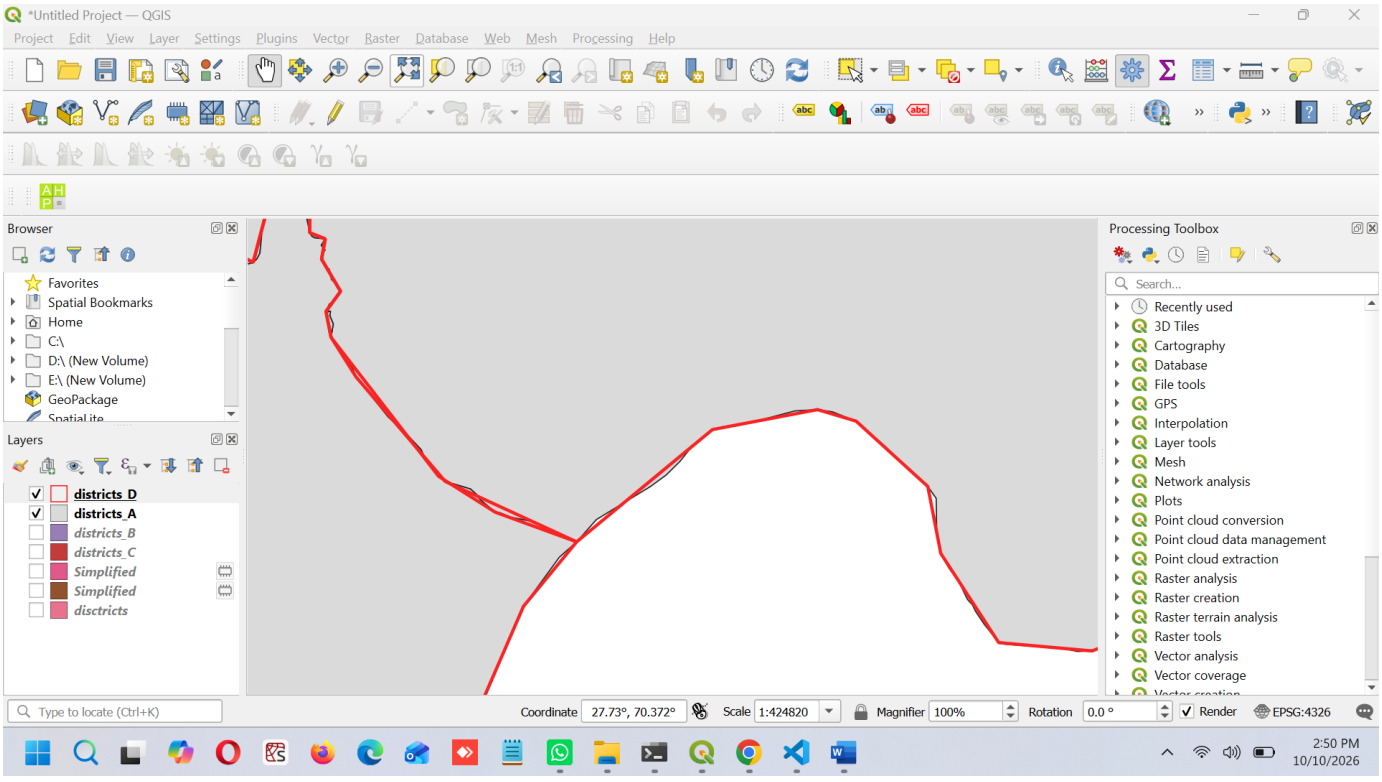
Checkpoint 1: Project Setup

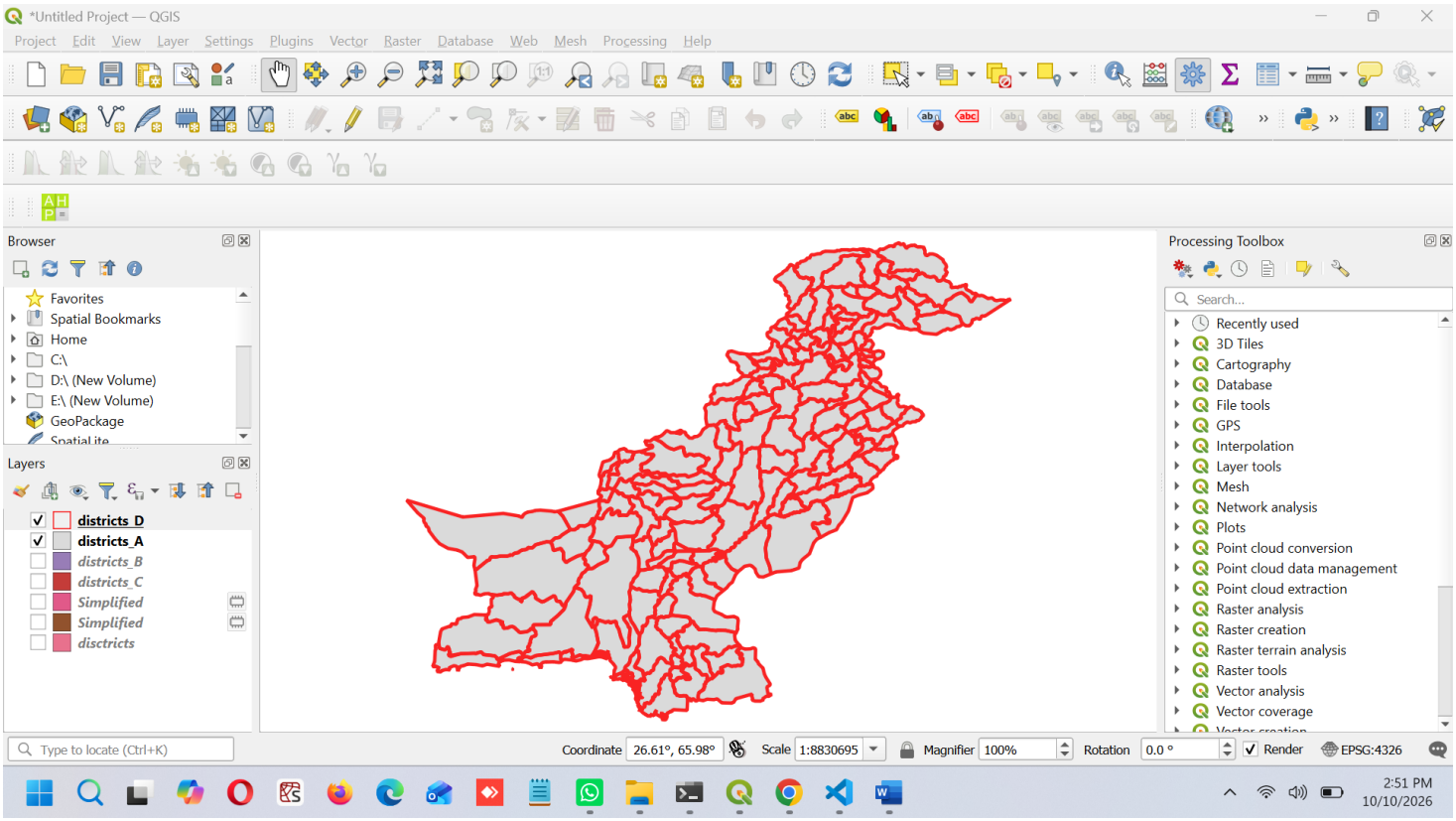
VS Code shows the folder C:\webgis-lab containing index.html, style.css, app.js, and an empty data folder. All three core files are created and ready.



Checkpoint 2: Data Preparation Measurements

File sizes recorded: Shapefile (24.3 MB), Version A (27.7 MB), B, C, and D decreasing with precision reduction and simplification. QGIS screenshot shows versions A and D overlaid, making the loss of detail visible





Checkpoint 3

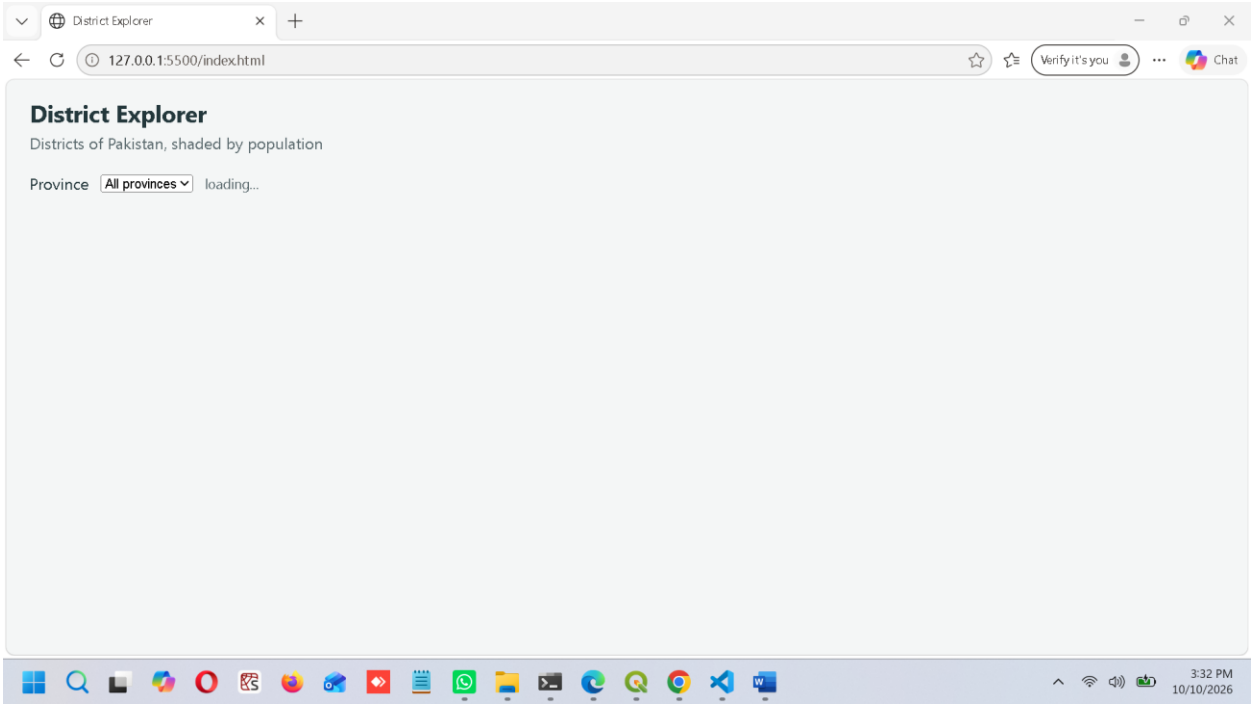
1. What is the type of the outermost object? = **FeatureCollection**
2. How many entries are in the features array? VS Code shows the count if you select the array. = **160**
3. Take the first feature. What is its geometry type?= **Polygon**
4. How many levels of square bracket are there before the first number in its coordinates? = **Three: [[[73.69622, 34.08025] ...**

   This is the structure for a Polygon: coordinates → rings → one coordinate pair.
5. List the keys inside its properties. = **district, province, pcode, pop_2023, area_km2**

First coordinate pair: [73.69622, 34.08025]

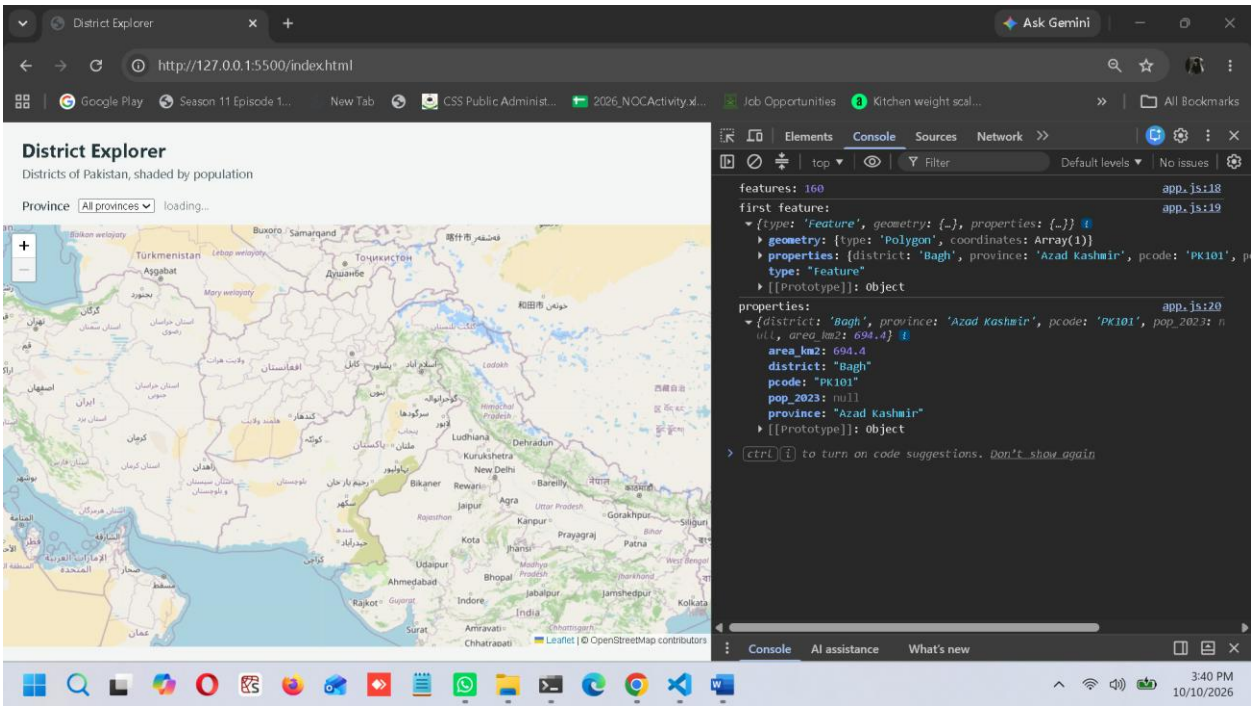
Checkpoint 4: Initial Page Load

Live Server page displays header, subtitle, province dropdown, and empty grey map area. Address bar confirms http://127.0.0.1:5500 rather than file://.



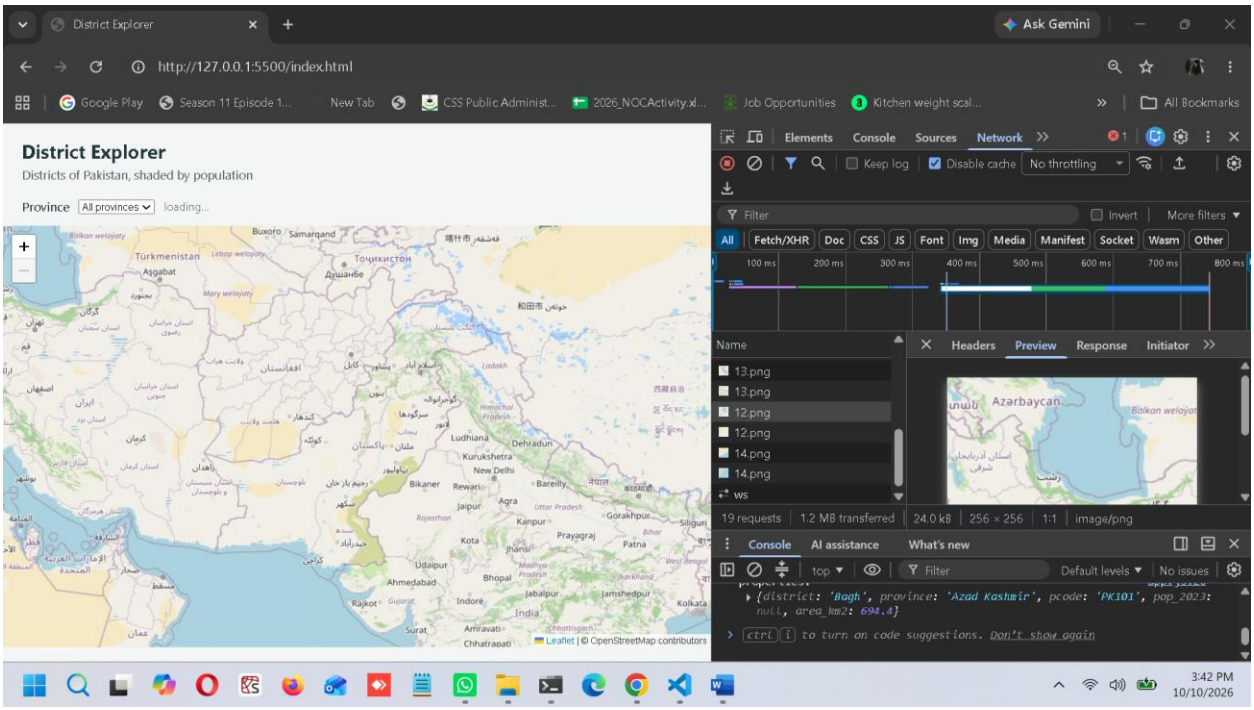
Checkpoint 5: Console Data Inspection

Console shows feature count expanded first feature with geometry and properties, and sorted list of distinct provinces. All array methods (length, map, filter, Set) return expected results.



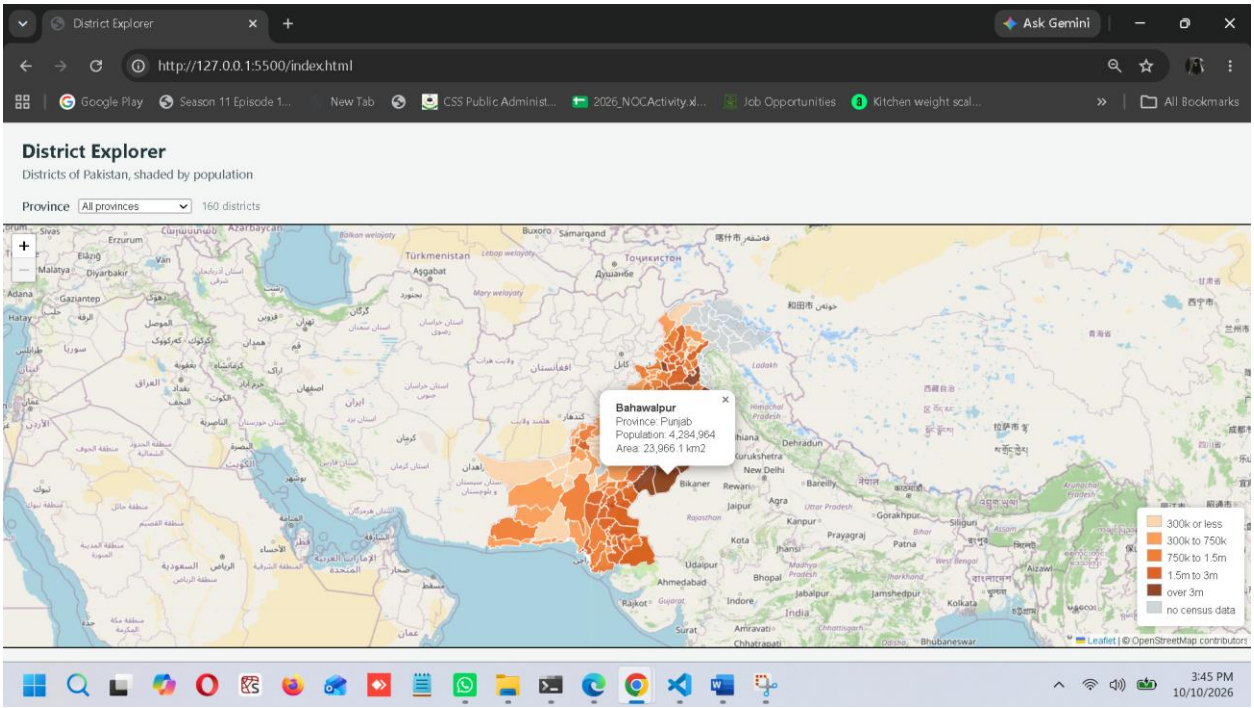
Checkpoint 6: Network Activity

Network tab shows the districts.geojson request with its size and load time, plus multiple tile .png requests. A selected tile preview reveals a single 256×256 pixel map square with zoom level in the URL.



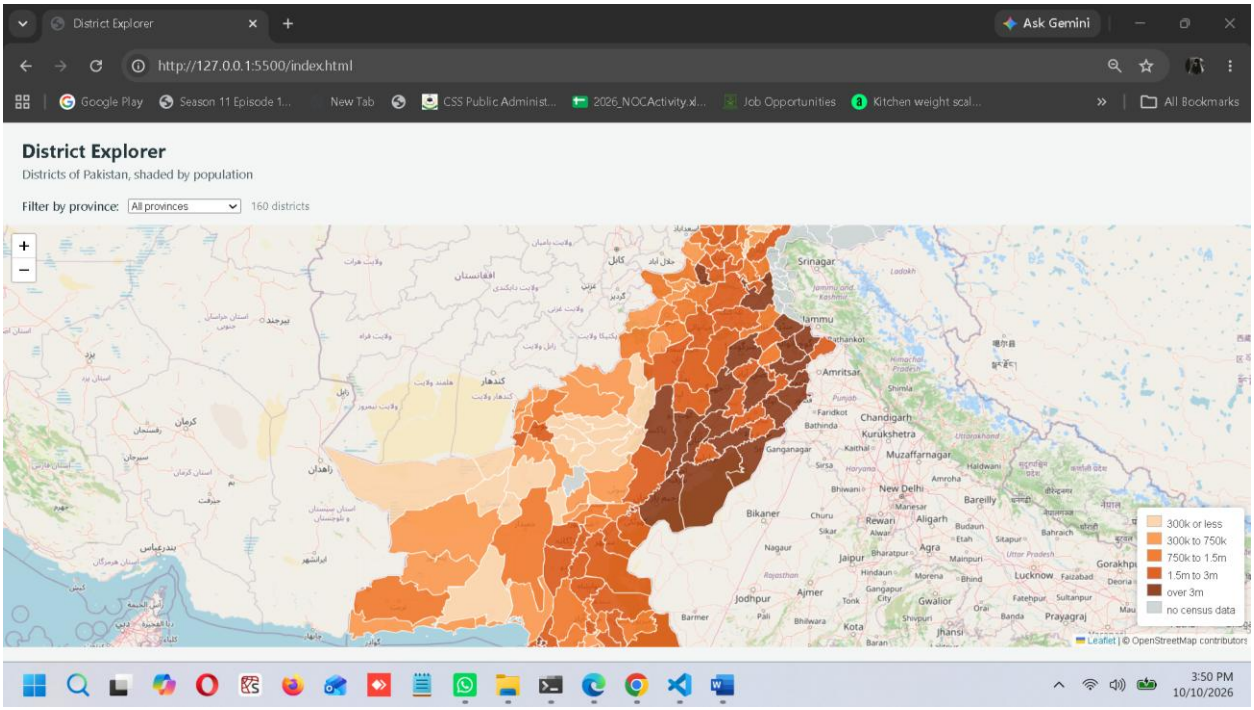
Checkpoint 7: Shaded Map with Legend and Popup

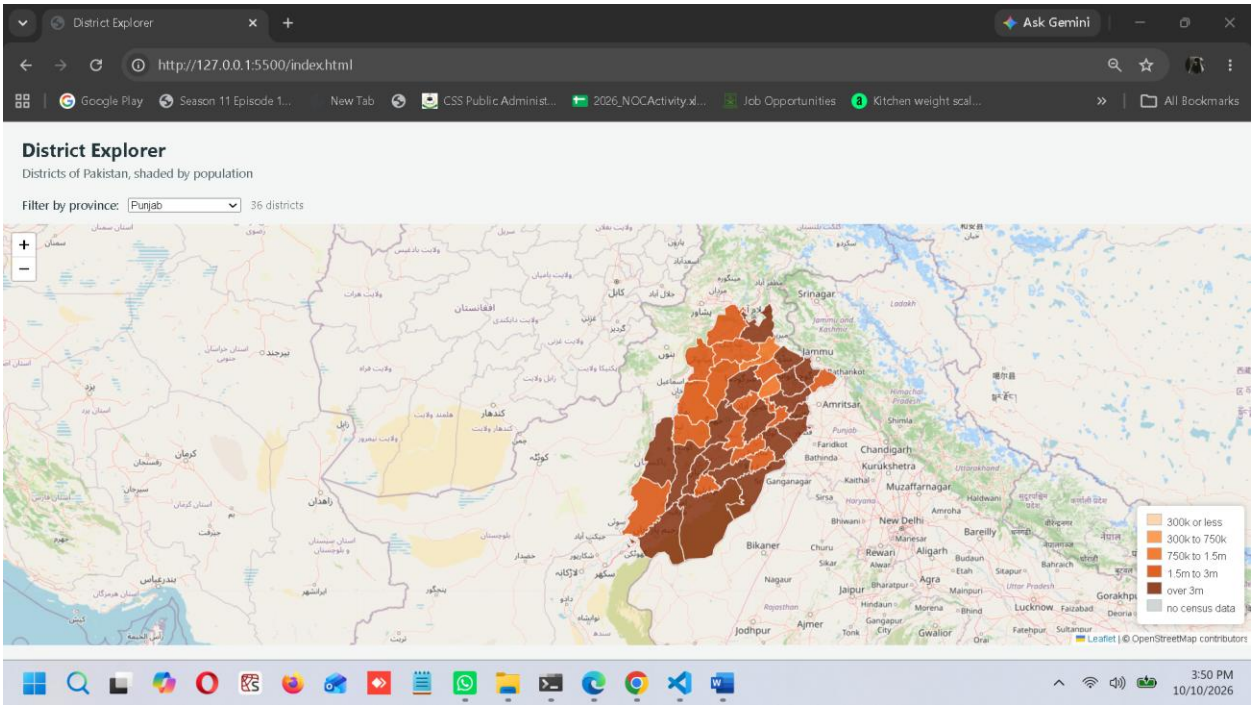
Map displays districts shaded by population using five classes plus grey for missing data. Legend visible at bottom-right, and a popup shows Bahawalpur's district, province, population, and area.



Checkpoint 8: Province Filter

Two screenshots: one showing all 160 districts across all provinces, and one showing only Punjab's 36 districts after dropdown selection. The district count updates correctly in both views.





Checkpoint 9:

District	Latitude (approx.)	3857 Area (km²)	UTM Area (km²)	Difference (%)
Thatta	24.7° N	9,290.174	7,725.939	20.24%
Dera Bugti	29.0° N	13,493.830	10,372.219	30.10%
Bahawalpur	29.4° N	31,346.462	24,010.576	30.55%
Bagh	33.9° N	1,012.646	694.047	45.91%
Shangla	34.9° N	2,060.243	1,384.641	48.79%